UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Particular Integral, Types 4 and 5: Exponential Shift and xV

Complete study pack for class 006: the exponential shift $\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$ with V a sine, cosine or polynomial (it hands the work back to Type 2 or Type 3), the rule $\frac{1}{f(D)}(xV) = \left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$, and every slide, class-work and ESA problem on them, solved in full and checked with sympy

Compiled from the PESU Unit 2 class 006 slide deck (15 slides), its AV-summary transcript (T006), the Type 4 part of the "Type 3 and Type 4" problem lecture (T006b), the Unit 1 & 2 class-work sheet (problems 17–20) and eight past ESA papers. Generated 16 October 2026.

· How to use this pack

Topic 1 is the Type 4 rule itself (Figure F1.1): pull e^{ax} to the left, replace D by $D + a$, then finish with the rule that fits V (Figure F1.2). Its examples have a sine or cosine V, including the failure case that the shift resolves on its own. Topic 2 is V a polynomial ($x^m e^{ax}$), which ends in the Type 3 series or in plain integration, and the ESA question (CW P18). Topic 3 is Type 5, $X = xV$, with the slide formula and, flagged as added for clarity, the shortcut $x \sin ax = \mathrm{Im}(xe^{iax})$. CF cases: class 002 pack; Type 1 (e^{ax}): class 003; Type 2 ($\sin$, $\cos$, $\sin^2 x$, rationalising, failure): class 004; Type 3 (polynomials): class 005. They are not re-taught here, but every worked example is complete. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S006·5	Unit 2 deck 006 (“Particular integral of Type 4”), slide 5 (rendered .pptx)
T006 00:04:12	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 4” (17 min), segment start
T006b 00:12:47	Unit 2 AV-summary transcript “Particular integral – Type 3 and Type 4” (bucket 006, 15.5 min); only its Type 4 part (00:09:55–00:15:25) is cited here. The class 005 pack cites the same file's Type 3 part as T006a.
CW U2 P18	Unit 2 class-work problem 18
ESA Jan-24 s3 2c	Past paper: UE23MA141A Jan 2024 set 3, question 2.c

Contents

1. 1 · The Type 4 rule: $X = e^{ax}V$ (exponential shift)
2. 2 · Type 4 with a polynomial: $x^m e^{ax}$ and mixed right-hand sides
3. 3 · The Type 5 rule: $X = xV$
4. Appendix A · Symbol glossary
5. Appendix B · Sources, coverage and gaps

1 The Type 4 rule: $X = e^{ax}V$ (exponential shift)

When the right-hand side is a product $e^{ax}V$, with V a function of x, the exponential can be pulled out of the inverse operator to the left, provided every D in $f(D)$ is replaced by $D + a$: $\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$. The new operator acts on V alone, so the problem drops back to a type already known: Type 2 when V is a sine or cosine, Type 3 when V is a polynomial, plain integration when $f(D + a)$ is a power of D. This topic takes $V = \sin bx$ or $\cos bx$.

Key terms and definitions

Type 4 right-hand side. $X = e^{ax}V$, where V is a function of x (in the problems: $\sin bx$, $\cos bx$, a polynomial, or a product such as $x\cos x$). S006·3 · S006·4 · S006·10 · T006 00:00:00

Exponential shift. Moving e^{ax} from the right of $\frac{1}{f(D)}$ to its left while replacing D by $D + a$: "e^{ax} can be taken outside the operator by replacing D by $D + a$ in $f(D)$". S006·4 · T006 00:01:00

a. The coefficient of x in the exponential factor: $a = 1$ for e^x, $a = 3$ for e^{3x}, $a = -1$ for e^{-x}. T006 00:01:18 · T006b 00:12:13

$f(D + a)$. The shifted operator: $f(D)$ with every D replaced by $D + a$, then expanded and simplified (e.g. $D^2 - 2D + 4 \to (D + 1)^2 - 2(D + 1) + 4 = D^2 + 3$). S006·6 · T006 00:03:41

> *"Type (iv); When $X = e^{ax}V$, where V is a function of x: $\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$."* S006·4

> *"When $X = e^{ax}V$, V being a function of x, the particular integral is given by $y_p = \text{P.I.} = \frac{1}{f(D)}e^{ax}V(x) = e^{ax}\frac{1}{f(D+a)}V(x) = e^{ax}[f(D + a)]^{-1}V(x)$ where $V(x)$ is of some particular form."* S006·10

Formulas

E1.1 · Type 4: the exponential shift ESA ★★

$$\frac{1}{f(D)}\left(e^{ax}V\right) = e^{ax}\,\frac{1}{f(D+a)}\,V = e^{ax}\left[f(D+a)\right]^{-1}V$$

gives	The PI of e^{ax} times any function V.
applies	$X = e^{ax}V$; a = coefficient of x in the exponent.
note	Write e^{ax} to the left first, then replace every D by $D + a$; V is untouched.

S006·4 · S006·10 · T006 00:00:52 · T006 00:01:00 · T006b 00:10:19 · ESA Jan-24 s3 2c

E1.2 · Why it works: D past e^{ax} ESA ★

$$D\big(e^{ax}V\big) = e^{ax}(D+a)V, \qquad f(D)\big(e^{ax}V\big) = e^{ax}f(D+a)V$$

GIVES The shift rule in the forward direction; inverting both sides gives E1.1.

APPLIES Product rule, repeated.

NOTE $D^2(e^{ax}V) = e^{ax}(D+a)^2V$, and so on for every power.

ADDED FOR CLARITY S006·4

E1.3 · Then $V = \sin bx$ or $\cos bx$ (Type 2) ESA ★★

$$e^{ax}\frac{1}{f(D+a)}\cos bx : \quad D^2 \to -b^2 \text{ in } f(D+a)$$

GIVES The second half of the work when V is trigonometric.

APPLIES After the shift; rationalise a leftover D (class 004).

NOTE Use b of V, not a of the exponential: $e^x\cos x$: $D^2 \to -1$.

S006·6 · T006 00:04:20 · T006b 00:10:53 · CW U2 P17

E1.4 · The Type 1 failure rule is a shift ESA ★

$$\frac{e^{ax}}{(D-a)^k\,\psi(D)} = e^{ax}\frac{1}{D^k\,\psi(D+a)}\,1 = \frac{x^k}{k!}\,\frac{e^{ax}}{\psi(a)}$$

GIVES Why a being a k-fold AE root brings in x^k.

APPLIES $V = 1$ and $f(a) = 0$ ($\psi(a) \neq 0$).

NOTE Same answer as $x\frac{e^{ax}}{f'(a)}$, $x^2\frac{e^{ax}}{f''(a)}$ of class 003.

ADDED FOR CLARITY S006·14

Graphs and diagrams

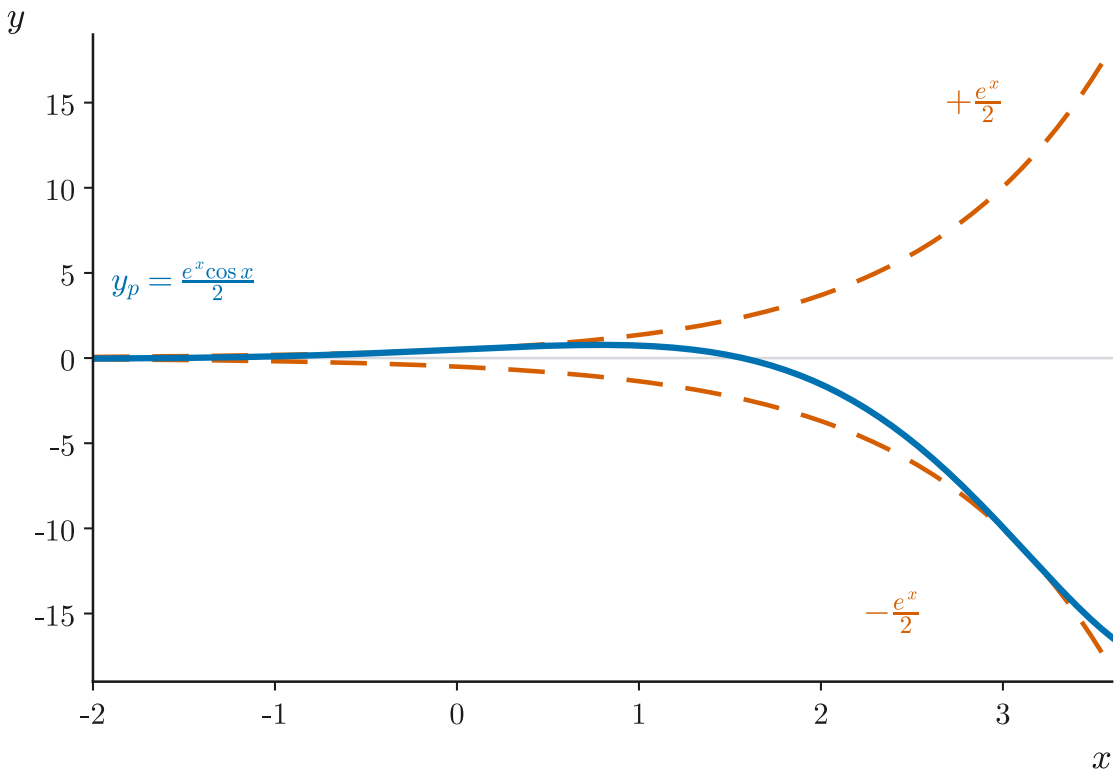
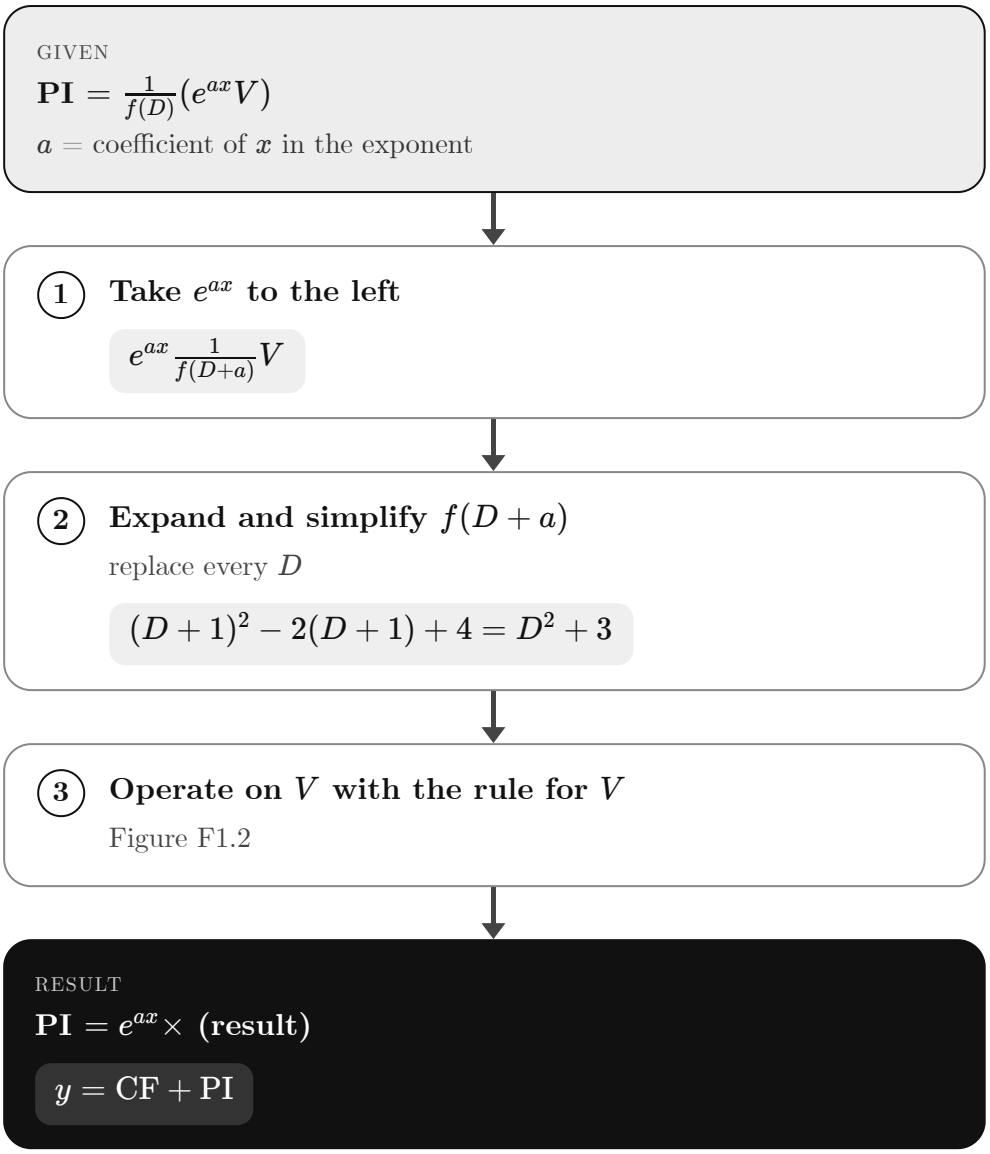


Figure G1.1. The PI $\frac{e^x \cos x}{2}$ oscillates inside its envelope $\pm\frac{e^x}{2}$. ESA ★★

ADDED FOR CLARITY S006·5 · S006·6

How to read it. The PI of $(D^2 - 2D + 4)y = e^x \cos x$ (blue) is the shifted problem's answer $\frac{\cos x}{2}$ multiplied by e^x: it touches the envelope $\pm\frac{e^x}{2}$ (vermilion dashed) where $\cos x = \pm 1$ and crosses zero where $\cos x = 0$. The factor e^{ax} survives the shift unchanged; only the oscillating part V is operated on.

Flowcharts

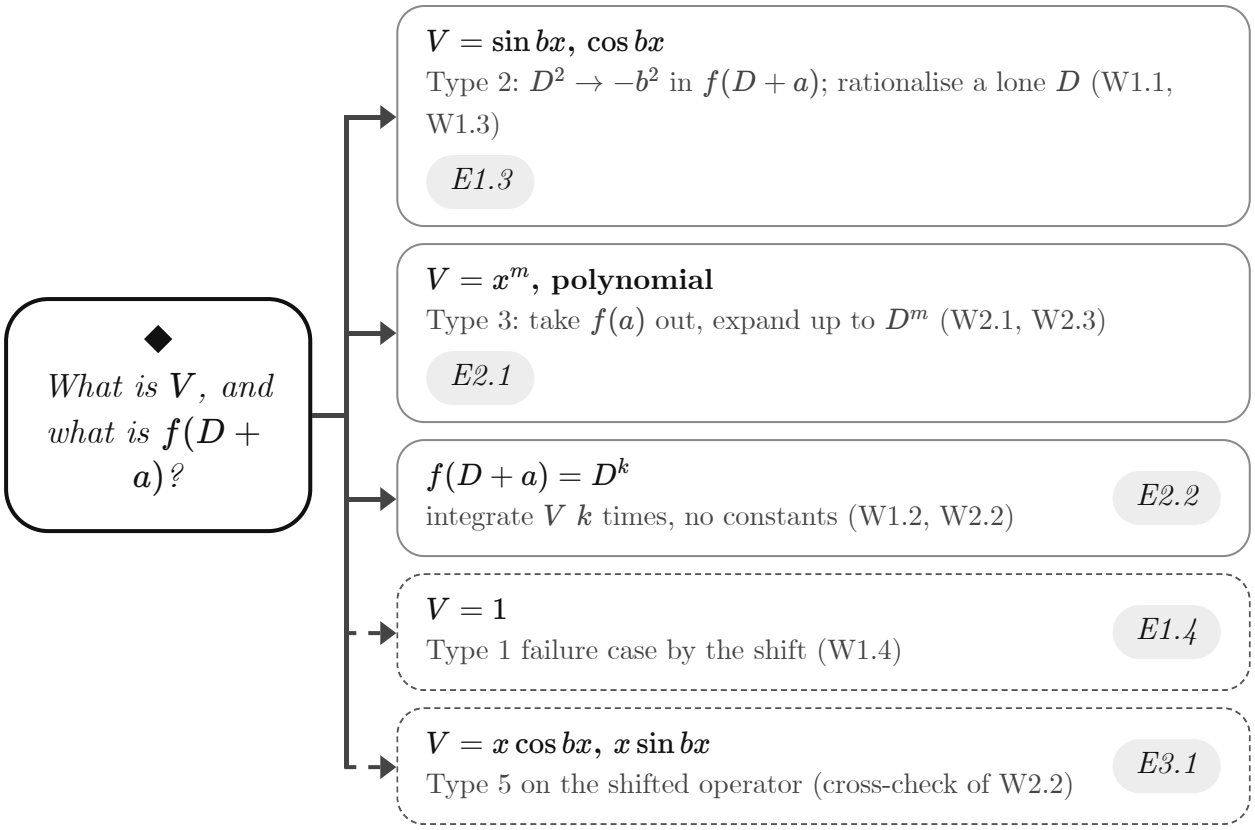


Remark. The exponential is never differentiated: it waits on the left while $f(D+a)$ works on V.

Exam trap. Replace the D in every term ($-2D \to -2(D+1)$), not only in D^2.

Figure F1.1. The Type 4 procedure (slide 006·4) ESA ★★

S006·4 · S006·10 · T006 00:01:00 · T006b 00:10:19 · T006b 00:10:53



Remark. T006b 00:10:53: "If V is sine or cos, we go with case 2. If V is a polynomial, we go with case 3."

Figure F1.2. After the shift: which rule for V? ESA ★★

T006b 00:10:53 · S006·6 · S006·8 · S006·12 · S006·14

The five PI types side by side

Type	X	Rule	Special case	Class pack
1	e^{ax}	$D \to a$	$f(a) = 0$: $x\frac{e^{ax}}{f'(a)}$	003
2	$\sin ax$, $\cos ax$	$D^2 \to -a^2$	$f(-a^2) = 0$: multiply by x, differentiate	004
3	x^m	$[f(D)]^{-1}$ in powers of D, up to D^m	$f(0) = 0$: take D^k out	005
4	$e^{ax}V$	$e^{ax}\frac{1}{f(D+a)}V$	none needed: the shift gives D^k	006 (this pack)
5	xV	$\left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$	use Type 2 on both terms	006 (this pack)

ADDED FOR CLARITY S006·4 · S006·10

Important points and why

1. Recognise Type 4: X is a product with an exponential factor, "e^{ax} into V" ($e^x \cos x$, $x^2 e^{3x}$, $e^{2x}\cos x$). "If x^2 is not there, case 1; if e^{3x} is not there, case 3; both are there, so case 4". ESA ★★ T006 00:00:18 · T006b 00:11:45 · S006·10
2. Write e^{ax} to the **left** of the operator and replace **every** D in $f(D)$ by $D+a$; leave V as it is. ESA ★★ S006·4 · T006 00:01:00 · T006b 00:10:19
3. Expand and simplify $f(D+a)$ before doing anything else: $(D+1)^2-2(D+1)+4 = D^2+3$; $(D+1)^2-2(D+1)+1 = D^2$. ESA ★★ S006·6 · S006·8 · T006 00:03:58 · T006 00:06:23
4. After the shift, the rule is chosen by V alone: $\sin/\cos \to$ Type 2; polynomial $\to$ Type 3; $f(D+a) = D^k \to$ integrate k times. ESA ★★ T006b 00:10:40 · T006b 00:10:53 · T006 00:04:20 · T006 00:12:09
5. In Type 2 after the shift, $D^2 \to -b^2$ uses b from $V = \cos bx$, not a from e^{ax}: $\frac{e^x \cos x}{D^2+3} \to \frac{e^x \cos x}{-1+3}$. ESA ★★ S006·6 · T006 00:04:20 · CW U2 P17
6. The shift handles failure by itself: if a is a root of the AE, $f(D+a)$ has a factor D, and $\frac{1}{D}$ integrates (slide 006·14: $(D-1)^2 \to D^2$). ESA ★★ S006·14 · T006b 00:14:22
7. "Find the general solution" needs $y = y_c + y_p$ written out: "this is must". ESA ★★ T006 00:09:28 · S006·6 · S006·9

Why does D become $D+a$?
By the product rule $D(e^{ax}V) = e^{ax}(DV + aV) = e^{ax}(D+a)V$; applying this repeatedly, $f(D)(e^{ax}V) = e^{ax}f(D+a)V$. Reading this backwards for the inverse operator gives E1.1: to undo $f(D)$ on $e^{ax}V$, undo $f(D+a)$ on V.
ADDED FOR CLARITY S006·4

Why does the shift resolve a failure case without a special rule?
If a is a k-fold root of the AE, $f(D) = (D-a)^k\psi(D)$, so $f(D+a) = D^k\psi(D+a)$: the shifted operator has D^k as a factor and the k integrations produce the extra x^k. Slide 006·14 is an example: $(D-1)^2 \to D^2$.
ADDED FOR CLARITY S006·14 · T006b 00:14:22

Why can Type 2 not be used directly on $e^x \cos x$?
Type 2 replaces D^2 by $-b^2$ because $D^2 \cos bx = -b^2 \cos bx$; but $D^2(e^x \cos x) = -2e^x \sin x$ is not a multiple of $e^x \cos x$. After the shift the operator acts on $\cos x$ alone, where the substitution is valid.
ADDED FOR CLARITY S006·6 · T006 00:03:23

Exam trap. Slide 006·5 writes the CF as $e^x(c_1 \cos\sqrt{3}x + c_1 \cos\sqrt{3}x)$; it is $e^x(c_1 \cos\sqrt{3}x + c_2 \sin\sqrt{3}x)$. ADDED FOR CLARITY S006·5 · T006 00:03:06

Exam trap. Do not leave e^{ax} inside: $\frac{1}{f(D+a)}(e^{ax}V)$ is wrong; the exponential moves to the left and disappears from what the operator acts on. S006·4 · T006 00:01:00

Exam trap. Replace every D, including the D inside $-2D$ and $4D$: $D^2+4D+3 \to (D+2)^2+4(D+2)+3$, not $(D+2)^2+4D+3$. ADDED FOR CLARITY CW U2 P17 · S006·6

Exam trap. Do not put $D=a$ (Type 1) when V is not a constant: that rule is only for $V=1$. ADDED FOR CLARITY S006·4

Worked examples

W1.1 · $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+4y=e^x\cos x$ (slides 006·5–6) ESA ★★

GIVEN Find the general solution of $y''-2y'+4y=e^x\cos x$, i.e. $(D^2-2D+4)y=e^x\cos x$.

STEP 1 AE $m^2-2m+4=0$: $m=1\pm\sqrt{3}i$ ($\alpha=1$, $\beta=\sqrt{3}$). CF $=e^x(c_1\cos\sqrt{3}x+c_2\sin\sqrt{3}x)$ (slide prints $c_1\cos$ twice).

STEP 2 PI $=\frac{e^x\cos x}{D^2-2D+4}=e^x\frac{\cos x}{(D+1)^2-2(D+1)+4}$ (shift, $a=1$).

STEP 3 $(D+1)^2-2(D+1)+4=D^2+2D+1-2D-2+4=D^2+3$.

STEP 4 Type 2: $D^2\to-1^2$: PI $=e^x\frac{\cos x}{-1+3}=\frac{e^x\cos x}{2}$.

ANSWER $\boxed{y=e^x(c_1\cos\sqrt{3}x+c_2\sin\sqrt{3}x)+\frac{e^x\cos x}{2}}$

CHECK PI agrees with the slide; sympy dsolve and checkodesol agree. Figure G1.1.

S006·5 · S006·6 · T006 00:01:55 · T006 00:04:34

W1.2 · $(D^2-2D+1)y=e^x\sin x$: the shift handles the repeated root (slides 006·13–14) ESA ★★

GIVEN Solve $(D^2-2D+1)y=e^x\sin x$.

STEP 1 AE $m^2-2m+1=0$: $m=1,1$. CF $=(c_1+c_2x)e^x$.

STEP 2 PI $=\frac{e^x\sin x}{(D-1)^2}=e^x\frac{\sin x}{(D+1-1)^2}=e^x\frac{1}{D^2}\sin x$.

STEP 3 Integrate twice: $\int\sin x\,dx=-\cos x$, $\int(-\cos x)\,dx=-\sin x$. (Or Type 2: $D^2\to-1$.)

ANSWER $\boxed{y=(c_1+c_2x)e^x-e^x\sin x}$

CHECK Slide answer agrees; sympy dsolve and checkodesol agree. $a=1$ is a double root, yet no failure rule was needed.

S006·13 · S006·14 · T006b 00:14:02 · T006b 00:15:02

W1.3 · $(D^2+4D+3)y = e^{2x}\cos x$ (CW P17) ESA ★★

GIVEN Solve $(D^2+4D+3)y = e^{2x}\cos x$.

STEP 1 AE $(m+1)(m+3)=0$: CF $= c_1e^{-x} + c_2e^{-3x}$.

STEP 2 PI $= e^{2x}\frac{\cos x}{(D+2)^2+4(D+2)+3} = e^{2x}\frac{\cos x}{D^2+8D+15}$.

STEP 3 $D^2 \to -1$: $e^{2x}\frac{\cos x}{8D+14} = e^{2x}\frac{(8D-14)\cos x}{64D^2-196} = e^{2x}\frac{(8D-14)\cos x}{-260}$.

STEP 4 $(8D-14)\cos x = -8\sin x - 14\cos x$: PI $= e^{2x}\frac{8\sin x+14\cos x}{260}$.

ANSWER $$\boxed{y = c_1e^{-x} + c_2e^{-3x} + \frac{e^{2x}(7\cos x + 4\sin x)}{130}}$$

CHECK CW answer agrees; sympy dsolve and checkodesol agree.

CW U2 P17

W1.4 · $(D-2)^2y = e^{2x}$: the Type 1 failure case by the shift ESA ★

GIVEN Find the PI of $(D-2)^2y = e^{2x}$ by the Type 4 rule ($V=1$).

STEP 1 $f(2)=0$, so Type 1 fails ($m=2$ is a double AE root).

STEP 2 Shift: $e^{2x}\frac{1}{(D+2-2)^2}1 = e^{2x}\frac{1}{D^2}1$.

STEP 3 Integrate twice: $\int 1\,dx = x$, $\int x\,dx = \frac{x^2}{2}$.

ANSWER $$\boxed{\text{PI} = \frac{x^2e^{2x}}{2}}$$

CHECK Same as the class 003 rule $x^2\frac{e^{2x}}{f''(2)} = \frac{x^2e^{2x}}{2}$; substituting back gives e^{2x} (sympy).

ADDED FOR CLARITY S006·14

SHORT-NOTE RECAP

Type 4: $X = e^{ax}V$. $\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$: take e^{ax} to the left, replace every D by $D+a$, simplify, then use the rule for V ($\sin/\cos$: $D^2 \to -b^2$). Reason: $D(e^{ax}V) = e^{ax}(D+a)V$. Examples: $\frac{e^x\cos x}{D^2-2D+4} = e^x\frac{\cos x}{D^2+3} = \frac{e^x\cos x}{2}$; $\frac{e^x\sin x}{(D-1)^2} = e^x\frac{\sin x}{D^2} = -e^x\sin x$; CW P17: $\frac{e^{2x}\cos x}{D^2+4D+3} = \frac{e^{2x}(7\cos x+4\sin x)}{130}$. A root a of the AE gives a factor D after the shift.

2 Type 4 with a polynomial: $x^m e^{ax}$ and mixed right-hand sides

When V is a polynomial (x^2e^{3x}, xe^{3x}, x^2e^{-x}), the shifted operator $\frac{1}{f(D+a)}$ is expanded in ascending powers of D exactly as in Type 3, up to D^m. If the shift leaves only D^k (because a is a k-fold root of the AE), $\frac{1}{D^k}$ is k integrations without constants. The one ESA question on Type 4 (Jan-24 set 3 2c, = CW P18) mixes x^2e^{-x} with $\sin^2 x$; each term gets its own rule and the PIs are added.

Key terms and definitions

$\frac{1}{D^k}$. k successive integrations with respect to x, "without writing the constant of integration". S006·8 · S006·9 · T006 00:06:58 · T006 00:07:16

Formulas

E2.1 · Then $V = x^m$ (Type 3) ESA ★★

$$\frac{1}{f(D)}\left(x^m e^{ax}\right) = \frac{e^{ax}}{b_0}\left[1 \pm \varphi(D)\right]^{-1} x^m, \quad f(D+a) = b_0\left[1 \pm \varphi(D)\right]$$

GIVES	The PI of a polynomial times an exponential.
APPLIES	After the shift, $f(D+a)$ has a constant term $b_0 = f(a) \neq 0$.
NOTE	Expand up to D^m only (class 005 series). The PI is e^{ax} times a polynomial of degree m.

S006·11 · S006·12 · T006b 00:12:47 · T006b 00:13:06 · T006 00:12:09 · CW U2 P18

E2.2 · When the shift leaves only D^k ESA ★★

$$\frac{1}{(D-a)^k}\left(e^{ax}V\right) = e^{ax}\frac{1}{D^k}V = e^{ax}\underbrace{\int\cdots\int}_{k} V\,dx\cdots dx$$

GIVES	The PI by k integrations (no constants).
APPLIES	$f(D) = (D-a)^k$, e.g. $D^2 - 2D + 1 = (D-1)^2$ with $a = 1$.
NOTE	Integrate by parts for $x\cos x$; "without writing the constant of integration, that is very important".

S006·8 · S006·9 · S006·14 · T006 00:06:40 · T006 00:07:16 · T006b 00:14:42

E2.3 · Integration by parts (for $\frac{1}{D}$ of $x\cos x$, $x\sin x$) ESA ★★

$$\int uv\,dx = u\int v\,dx - \int \Big(\int v\,dx\Big)u'\,dx$$

GIVES $\int x\cos x\,dx = x\sin x + \cos x$, $\int x\sin x\,dx = -x\cos x + \sin x$.

APPLIES $\frac{1}{D}$ of x times a sine or cosine.

NOTE Slide 006·9 integrates $x\sin x$ instead of $x\cos x$ (gap list).

S006·9 · T006 00:07:34 · T006 00:09:11

E2.4 · Superposition ESA ★★

$$\frac{1}{f(D)}(X_1 + X_2) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2 = \mathrm{PI}_1 + \mathrm{PI}_2$$

GIVES The PI of a mixed right-hand side.

APPLIES X = a Type 4 term plus a Type 1, 2 or 3 term.

NOTE Each term chooses its own rule; see the class 005 pack for more mixed questions.

T006 00:10:40 · ESA Jan-24 s3 2c · CW U2 P18

E2.5 · Reminder: $\sin^2 x$ and a factor D ESA ★★

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \frac{1}{D\,\psi(D)}\,k = \frac{kx}{\psi(0)}$$

GIVES The constant part of $\sin^2 x$ when D is a factor of $f(D)$.

APPLIES ESA Jan-24 s3 2c: $f(D) = D(D-1)^2$.

NOTE Taught in the class 004 and 005 packs.

ESA Jan-24 s3 2c · CW U2 P18

Graphs and diagrams

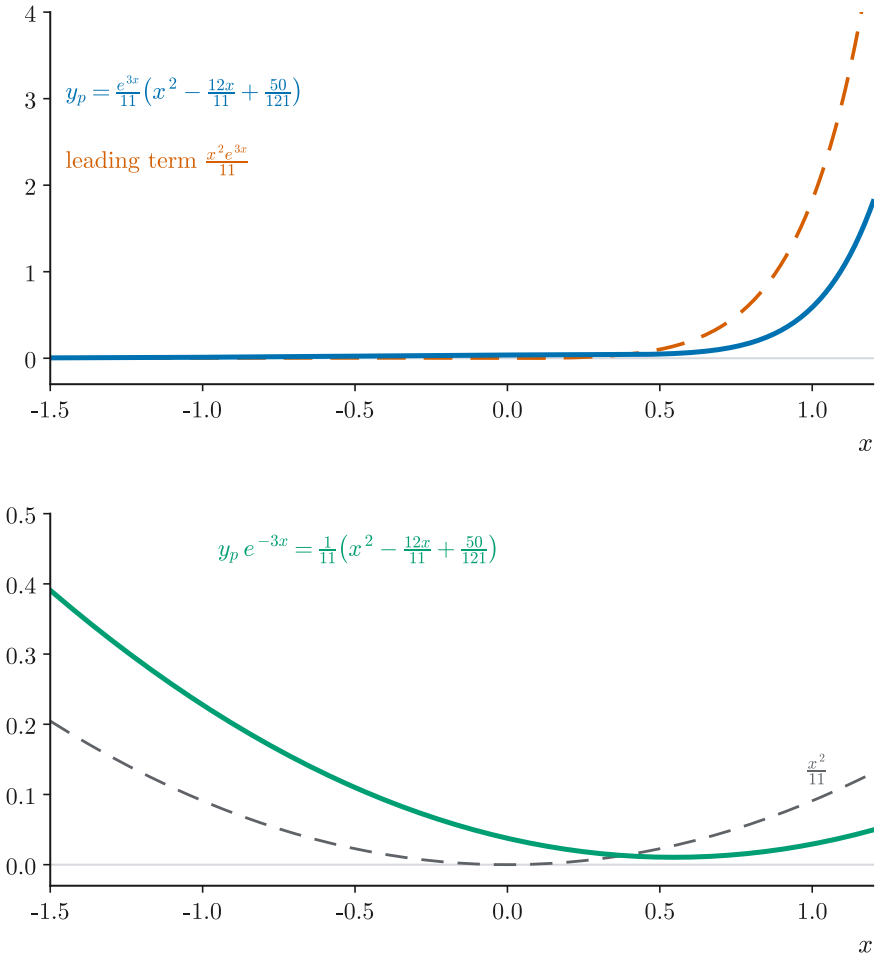
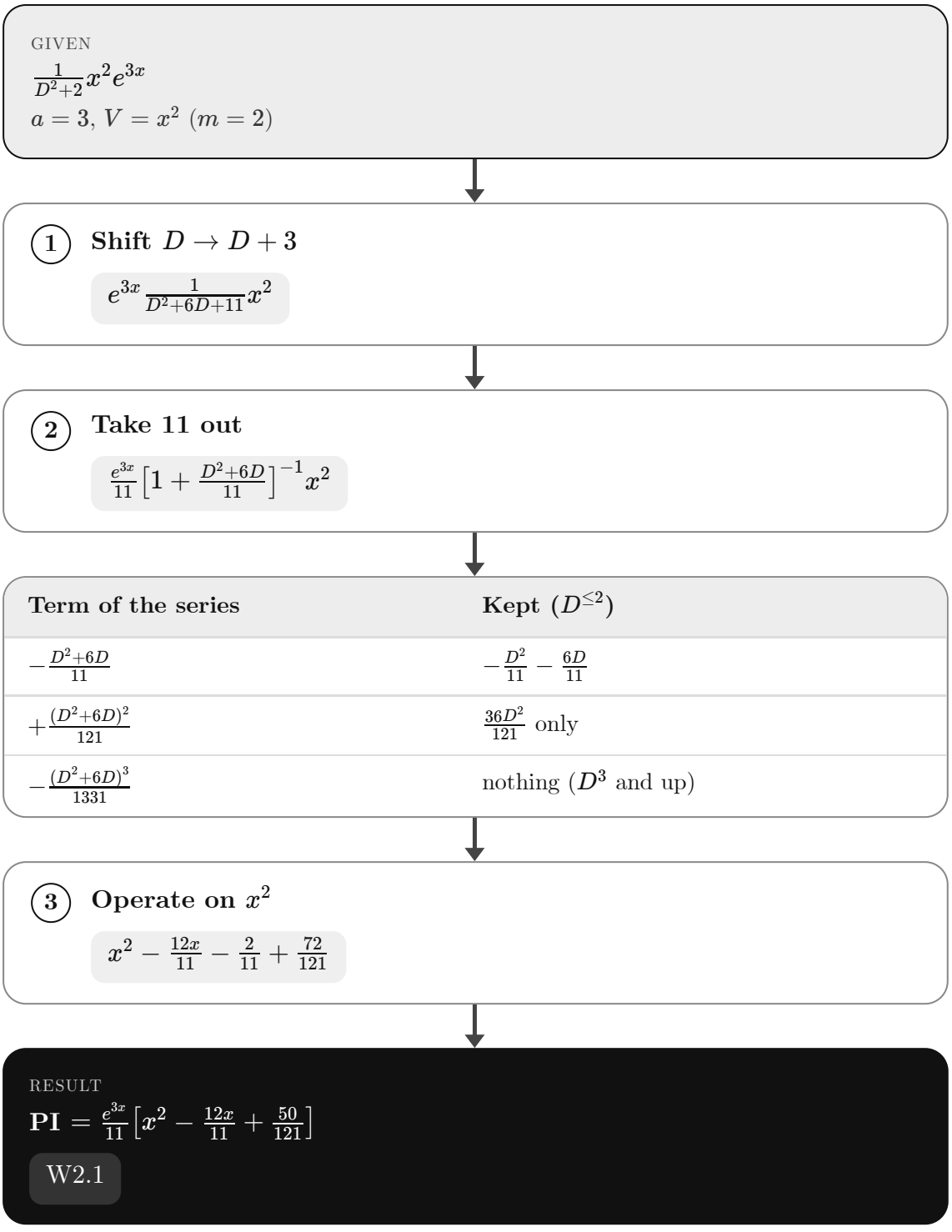


Figure G2.1. Strip the exponential: $y_p e^{-3x}$ is the Type 3 polynomial. ESA ★★

ADDED FOR CLARITY S006·11 · S006·12

How to read it. Top: the PI $y_p = \frac{e^{3x}}{11}\left(x^2 - \frac{12x}{11} + \frac{50}{121}\right)$ of $(D^2+2)y = x^2 e^{3x}$ (blue) next to $\frac{x^2 e^{3x}}{11}$ (vermilion dashed), the leading term alone. Bottom: divided by e^{3x}, the PI is the polynomial $\frac{1}{11}\left(x^2 - \frac{12x}{11} + \frac{50}{121}\right)$ (green), the answer of the shifted Type 3 problem $\frac{1}{D^2+6D+11}x^2$; the correction terms $-\frac{12x}{121} + \frac{50}{1331}$ move its vertex off the origin.

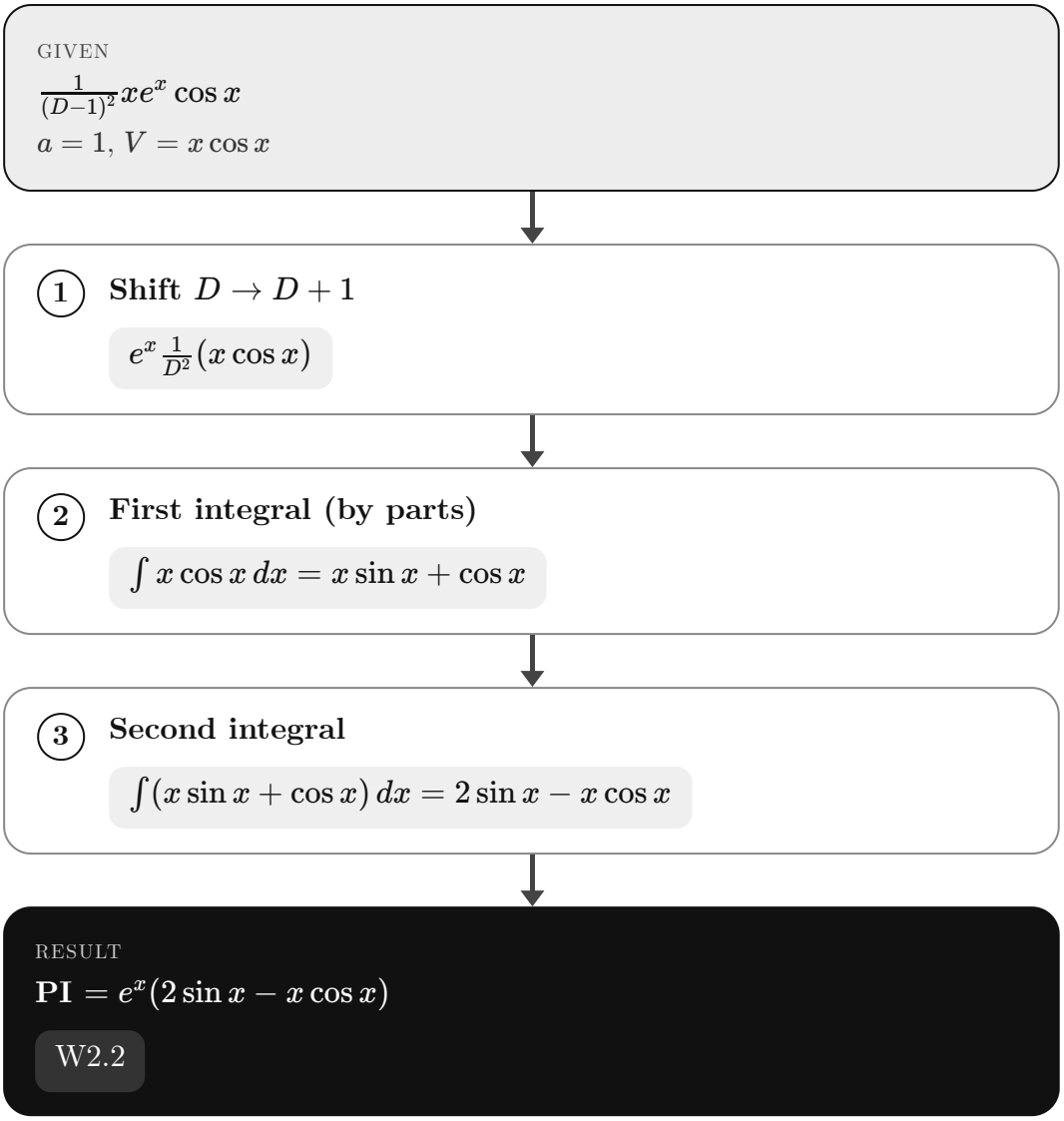
Flowcharts



Remark. "a^2 is not needed, b^2 is needed, $2ab$ is not needed" (T006b 00:13:25).

Figure F2.1. Slides 006·11–12: $(D^2+2)y=x^2e^{3x}$ ESA ★★

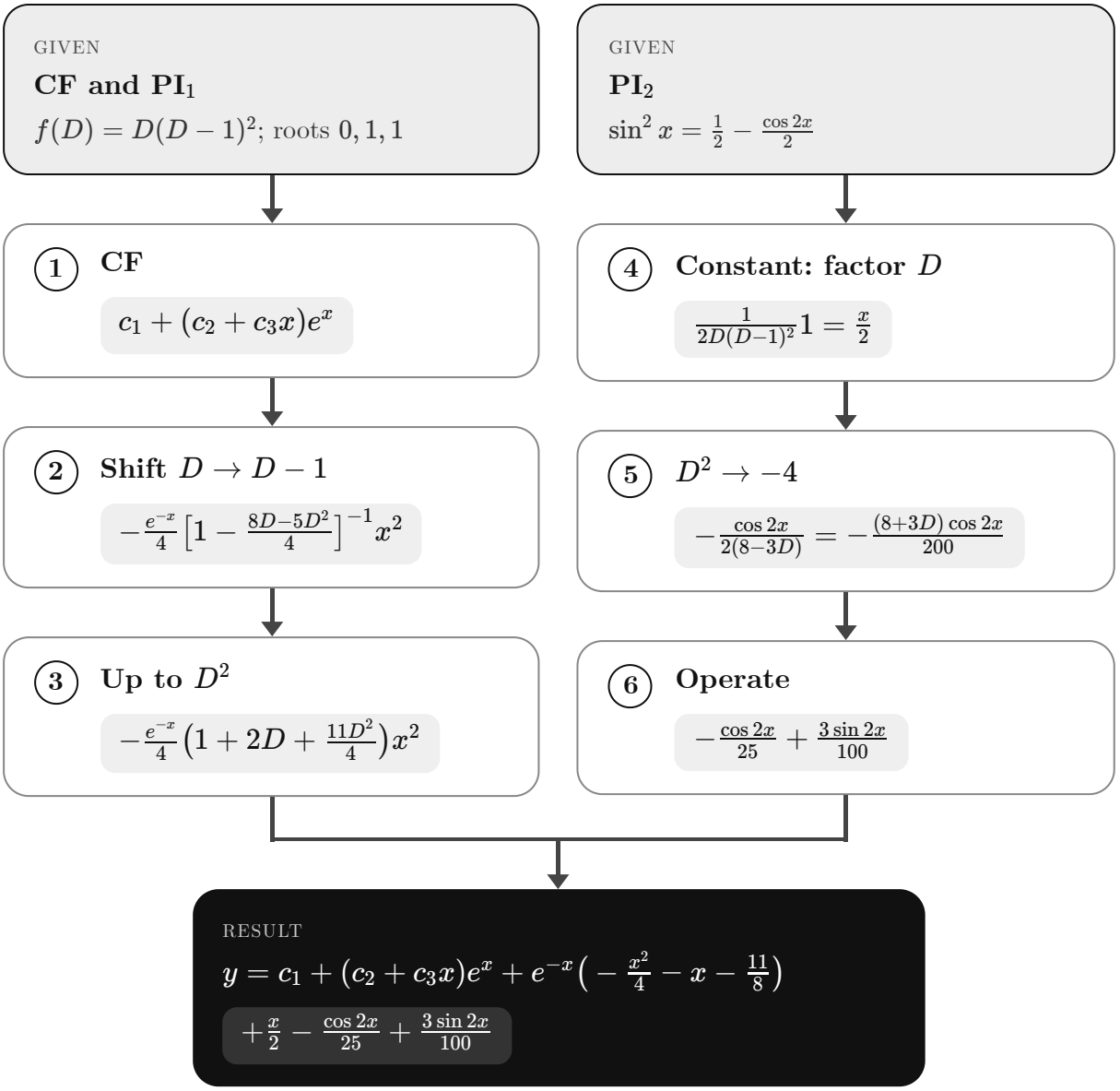
S006·11 · S006·12 · T006b 00:12:13 · T006b 00:13:25



Exam trap. Slide 006·9 integrates $x\sin x$ at step 2 and ends with $-e^x(x\sin x + 2\cos x)$, which does not satisfy the equation.

Figure F2.2. Slides 006·7–9: $(D-1)^2 y = xe^x \cos x$, corrected ESA ★★

S006·8 · S006·9 · T006 00:06:40 · T006 00:07:16



Remark. Worked in full as W2.3 (6 marks). The $\sin^2 x$ part is the class 004 pack's W3.2.

Figure F2.3. ESA Jan-24 s3 2c (= CW P18): $(D^3 - 2D^2 + D)y = x^2 e^{-x} + \sin^2 x$
ESA ★★

ESA Jan-24 s3 2c · CW U2 P18

Every Type 4 and Type 5 problem in this pack

Source (worked)	Equation	After the shift / rule	PI
S006·5–6 (W1.1)	$(D^2 - 2D + 4)y = e^x \cos x$	$e^x \frac{\cos x}{D^2+3}$	$\frac{e^x \cos x}{2}$
S006·13–14 (W1.2)	$(D-1)^2 y = e^x \sin x$	$e^x \frac{\sin x}{D^2}$	$-e^x \sin x$

Source (worked)	Equation	After the shift / rule	PI
CW P17 (W1.3)	$(D^2+4D+3)y = e^{2x}\cos x$	$e^{2x}\frac{\cos x}{D^2+8D+15}$	$\frac{e^{2x}(7\cos x+4\sin x)}{130}$
S006·11–12 (W2.1)	$(D^2+2)y = x^2e^{3x}$	$e^{3x}\frac{x^2}{D^2+6D+11}$	$\frac{e^{3x}}{11}(x^2-\frac{12x}{11}+\frac{50}{121})$
S006·7–9 (W2.2)	$(D-1)^2y = xe^x\cos x$	$e^x\frac{x\cos x}{D^2}$	$e^x(2\sin x - x\cos x)$
ESA s3 2c = P18 (W2.3)	$(D^3-2D^2+D)y = x^2e^{-x}+\dots$	$e^{-x}\frac{x^2}{D^3-5D^2+8D-4}$	$e^{-x}(-\frac{x^2}{4}-x-\frac{11}{8})+\dots$
T006 only (W2.4)	$(D^2-3D+2)y = xe^{3x}+\sin 2x$	$e^{3x}\frac{x}{D^2+3D+2}$	$\frac{e^{3x}}{2}(x-\frac{3}{2})+\frac{3\cos 2x-\sin 2x}{20}$
CW P19 (W3.1)	$(D^2-1)y = x\sin 3x$	Type 5	$-\frac{x\sin 3x}{10}-\frac{3\cos 3x}{50}$
CW P20 (W3.2)	$(D^2-5D+6)y = x\cos 2x$	Type 5	$\frac{x(\cos 2x-5\sin 2x)}{52}-\frac{73\sin 2x}{1352}-\frac{5\cos 2x}{169}$

ADDED FOR CLARITY

S006·5 · S006·7 · S006·11 · S006·13 · T006 00:09:45 · CW U2 P17 · CW U2 P18 · CW U2 P19 · CW U2 P20 · ESA Jan-24 s3 2c

Important points and why

1. For x^me^{ax}, take the constant $f(a)$ out of $f(D+a)$ and expand $[1\pm\varphi(D)]^{-1}$ up to D^m (class 005 rule). ESA ★★ S006·11 · S006·12 · T006b 00:12:47 · T006 00:12:25
2. Stop the series at the degree of the polynomial: for x^2 keep D^2 ("a^2 is not needed, b^2 is needed, $2ab$ is not needed"); for x keep only the D term. ESA ★★ S006·12 · T006b 00:13:25 · T006 00:13:20
3. $\frac{1}{D^2}$ means integrate twice, with no constants (the constants are already in the CF). ESA ★★ S006·8 · T006 00:06:58 · T006 00:07:16
4. With a mixed X (ESA Jan-24 s3 2c), split X, find PI_1 by the shift and PI_2 by Type 2, and add. ESA ★★ ESA Jan-24 s3 2c · CW U2 P18 · T006 00:10:40
5. Leading-term check: the PI of x^me^{ax} starts with $\frac{x^me^{ax}}{f(a)}$ when $f(a)\neq 0$ ($\frac{x^2e^{3x}}{11}$, $-\frac{x^2e^{-x}}{4}$). ESA ★ ADDED FOR CLARITY S006·12 · CW U2 P18

Exam trap. Slide 006·9 (and T006 00:07:16) integrates $x\sin x$ instead of $x\cos x$; its answer $-e^x(x\sin x+2\cos x)$ is wrong. $\frac{1}{D^2}(x\cos x)=2\sin x-x\cos x$, so the PI is $e^x(2\sin x-x\cos x)$. ADDED FOR CLARITY S006·9 · T006 00:07:16

Exam trap. No constants of integration when evaluating $\frac{1}{D}$: "that's very important". T006 00:07:16

Exam trap. Slide 004·17's boxed PI for the $\sin^2 x$ part of ESA Jan-24 s3 2c drops a factor 2; the correct part is $\frac{x}{2}-\frac{\cos 2x}{25}+\frac{3\sin 2x}{100}$ (class 004 pack, CW P18). ADDED FOR CLARITY ESA Jan-24 s3 2c · CW U2 P18

Worked examples

W2.1 · $(D^2+2)y = x^2e^{3x}$ (slides 006·11–12) ESA ★★

GIVEN Solve $(D^2+2)y = x^2e^{3x}$.

STEP 1 AE $m^2+2=0$: $m=\pm\sqrt{2}i$. CF $= c_1\cos\sqrt{2}x + c_2\sin\sqrt{2}x$.

STEP 2 PI $= e^{3x}\frac{1}{(D+3)^2+2}x^2 = e^{3x}\frac{1}{D^2+6D+11}x^2 = \frac{e^{3x}}{11}\left[1+\frac{D^2+6D}{11}\right]^{-1}x^2$.

STEP 3 Up to D^2: $1 - \frac{D^2+6D}{11} + \frac{36D^2}{121}$ (from the square keep only $36D^2$; the cube has D^3 and higher).

STEP 4 $D(x^2)=2x$, $D^2(x^2)=2$: $\frac{e^{3x}}{11}\left[x^2 - \frac{2}{11} - \frac{12x}{11} + \frac{72}{121}\right]$.

ANSWER $$\boxed{y = c_1\cos\sqrt{2}x + c_2\sin\sqrt{2}x + \frac{e^{3x}}{11}\left[x^2 - \frac{12x}{11} + \frac{50}{121}\right]}$$

CHECK Slide answer agrees; sympy dsolve and checkodesol agree. Figure G2.1.

S006·11 · S006·12 · T006b 00:11:07 · T006b 00:13:44

W2.2 · $y'' - 2y' + y = xe^x\cos x$ (slides 006·7–9, corrected) ESA ★★

GIVEN Find the general solution of $y'' - 2y' + y = xe^x\cos x$, i.e. $(D^2-2D+1)y = xe^x\cos x$.

STEP 1 AE $m^2-2m+1=0$: $m=1,1$. CF $=(c_1+c_2x)e^x$.

STEP 2 PI $= e^x\frac{x\cos x}{(D+1)^2-2(D+1)+1} = e^x\frac{1}{D^2}(x\cos x)$ ($V = x\cos x$, $a=1$).

STEP 3 First $\frac{1}{D}$: $\int x\cos x\,dx = x\sin x + \cos x$ (by parts). **The slide integrates $x\sin x$ here.**

STEP 4 Second $\frac{1}{D}$: $\int(x\sin x + \cos x)\,dx = (-x\cos x + \sin x) + \sin x = 2\sin x - x\cos x$.

STEP 5 PI $= e^x(2\sin x - x\cos x)$.

ANSWER $$\boxed{y = (c_1+c_2x)e^x + e^x(2\sin x - x\cos x)}$$

CHECK sympy dsolve and checkodesol agree. The slide's $-e^x(x\sin x + 2\cos x)$ fails the substitution test. Cross-check by Type 5 with $f(D)=D^2$: $x\frac{\cos x}{D^2} - \frac{2D}{D^4}\cos x = -x\cos x + 2\sin x$.

S006·7 · S006·8 · S006·9 · T006 00:04:55 · T006 00:09:11

W2.3 · $(D^3-2D^2+D)y = x^2e^{-x}+\sin^2 x$ (ESA Jan-24 s3 2c; CW P18) ESA ★★

GIVEN Solve $(D^3-2D^2+D)y = x^2e^{-x}+\sin^2 x$.

STEP 1 $f(D) = D(D-1)^2$; AE roots $0,1,1$: CF $= c_1 + (c_2+c_3x)e^x$.

STEP 2 PI_1: shift $a=-1$: $f(D-1) = (D-1)(D-2)^2 = D^3-5D^2+8D-4 = -4\big[1 - \frac{8D-5D^2+D^3}{4}\big]$.

STEP 3 $(1-z)^{-1}$ up to D^2: $1+\big(2D-\frac{5D^2}{4}\big)+4D^2 = 1+2D+\frac{11D^2}{4}$; on x^2: $x^2+4x+\frac{11}{2}$.

STEP 4 $\text{PI}_1 = -\frac{e^{-x}}{4}\big(x^2+4x+\frac{11}{2}\big) = e^{-x}\big(-\frac{x^2}{4}-x-\frac{11}{8}\big)$.

STEP 5 PI_2: $\sin^2 x = \frac12 - \frac{\cos 2x}{2}$. $\frac{1}{2f(D)} = \frac{1}{2D(D-1)^2}1 = \frac{x}{2}$ (factor D: integrate once).

STEP 6 $-\frac{\cos 2x}{2f(D)}$: $D^2\to -4$: $f \to -4D+8+D = 8-3D$; $-\frac{(8+3D)\cos 2x}{2(64-9D^2)} = -\frac{8\cos 2x - 6\sin 2x}{200}$.

ANSWER
$$\boxed{y = c_1 + (c_2+c_3x)e^x + e^{-x}\Big(-\frac{x^2}{4}-x-\frac{11}{8}\Big) + \frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}}$$

CHECK CW answer agrees ($\frac12(x - \frac{1}{100}(-6\sin 2x + 8\cos 2x))$ is the same); sympy dsolve and checkodesol agree. Slide 004·17's boxed $\sin^2 x$ part drops a factor 2.

ESA Jan-24 s3 2c · CW U2 P18

W2.4 · $(D^2-3D+2)y = xe^{3x}+\sin 2x$ (lecture example) ESA ★★

GIVEN Find the general solution of $y''-3y'+2y = xe^{3x}+\sin 2x$.

STEP 1 AE $m^2-3m+2=0$: $m=1,2$. CF $= c_1e^x + c_2e^{2x}$.

STEP 2 $\text{PI}_1 = e^{3x}\frac{x}{(D+3)^2-3(D+3)+2} = e^{3x}\frac{x}{D^2+3D+2} = \frac{e^{3x}}{2}\big[1+\frac{D^2+3D}{2}\big]^{-1}x$.

STEP 3 $m=1$: keep $1-\frac{3D}{2}$: $\text{PI}_1 = \frac{e^{3x}}{2}\big(x-\frac32\big)$.

STEP 4 PI_2: $D^2\to -4$: $\frac{\sin 2x}{-2-3D} = -\frac{(3D-2)\sin 2x}{9D^2-4} = \frac{(3D-2)\sin 2x}{40} = \frac{6\cos 2x - 2\sin 2x}{40}$.

ANSWER
$$\boxed{y = c_1e^x + c_2e^{2x} + \frac{e^{3x}}{2}\Big(x-\frac32\Big) + \frac{3\cos 2x - \sin 2x}{20}}$$

CHECK sympy dsolve and checkodesol agree. The lecture stops at "PI = PI_1+PI_2".

TRANSCRIPT ONLY T006 00:09:45 · T006 00:14:10 · T006 00:16:19

Past-paper questions

ESA Jan 2024 · 6 MARKS · UE23MA141A set 3, Q2.c (= CW P18) · solved in W2.3

Solve $(D^3-2D^2+D)y = x^2e^{-x}+\sin^2 x$.

SHORT-NOTE RECAP

V a polynomial: $\frac{x^m e^{ax}}{f(D)} = e^{ax}[f(D+a)]^{-1}x^m$, take $f(a)$ out, expand up to D^m. $\frac{x^2e^{3x}}{D^2+2} = \frac{e^{3x}}{11}(x^2 - \frac{12x}{11} + \frac{50}{121})$. If $f(D+a)=D^k$, integrate k times, no constants: $\frac{xe^x\cos x}{(D-1)^2} = e^x(2\sin x - x\cos x)$ (slide 006·9 is wrong). ESA: $\frac{x^2e^{-x}+\sin^2 x}{D^3-2D^2+D} = e^{-x}(-\frac{x^2}{4}-x-\frac{11}{8}) + \frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$.

3 The Type 5 rule: $X = xV$

When X is x times a function V (in practice $x\sin ax$ or $x\cos ax$), slide 006·4 gives $\frac{1}{f(D)}(xV) = \left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$: find $\frac{1}{f(D)}V$ and multiply it by x, then subtract $\frac{f'(D)}{(f(D))^2}V$, both by the Type 2 rule. The deck states the rule but works no example; the class-work sheet has two (P19, P20). A shorter route, added for clarity, writes $x\sin ax = \mathrm{Im}(xe^{iax})$ and uses Type 4 with $V = x$.

Key terms and definitions

Type 5 right-hand side. $X = xV$, where V is a function of x (in the problems: $\sin ax$ or $\cos ax$). S006·4 · T006 00:01:28

$f'(D)$. The derivative of the polynomial $f(D)$ with respect to D: for $f(D) = D^2 - 5D + 6$, $f'(D) = 2D - 5$. S006·4 · T006 00:01:28

> *"Type (v); When $X = xV$ where V is a function of x: $\frac{1}{f(D)}(xV) = \left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$."* S006·4

Formulas

E3.1 · Type 5: $X = xV$ ESA ★★

$$\frac{1}{f(D)}(xV) = \left[\frac{x}{f(D)} - \frac{f'(D)}{\left(f(D)\right)^2}\right]V = x\,\frac{1}{f(D)}V - \frac{f'(D)}{\left(f(D)\right)^2}V$$

gives	The PI of x times a function (usually $x\sin ax$, $x\cos ax$).
applies	$X = xV$; $f'(D) = \frac{d}{dD}f(D)$.
note	$\frac{x}{f(D)}V$ means: apply $\frac{1}{f(D)}$ to V, then multiply by x.

S006·4 · T006 00:01:28 · CW U2 P19 · CW U2 P20

E3.2 · Where Type 5 comes from ESA ★

$$f(D)(xU) = x\,f(D)U + f'(D)U \;\Rightarrow\; \frac{1}{f(D)}(xV) = x\frac{1}{f(D)}V - \frac{1}{f(D)}f'(D)\frac{1}{f(D)}V$$

gives	E3.1, with $U = \frac{1}{f(D)}V$; constant-coefficient operators commute, so the last term is $\frac{f'(D)}{(f(D))^2}V$.
applies	Leibniz: $D^n(xU) = xD^nU + nD^{n-1}U$.
note	Not on the slides.

ADDED FOR CLARITY S006·4

E3.3 · Shortcut: $x\sin ax$ as an imaginary part ESA ★

$$\frac{1}{f(D)}\,x\sin ax=\operatorname{Im}\Big[e^{iax}\frac{1}{f(D+ia)}\,x\Big],\qquad \frac{1}{f(D)}\,x\cos ax=\operatorname{Re}\Big[\,\dots\Big]$$

GIVES The same PI by Type 4 (with $a\to ia$) and Type 3 ($V=x$).

APPLIES $f(D)$ has real coefficients.

NOTE Only for checking or for fast work; the slide method is E3.1.

ADDED FOR CLARITY S006·4 · CW U2 P19 · CW U2 P20

Graphs and diagrams

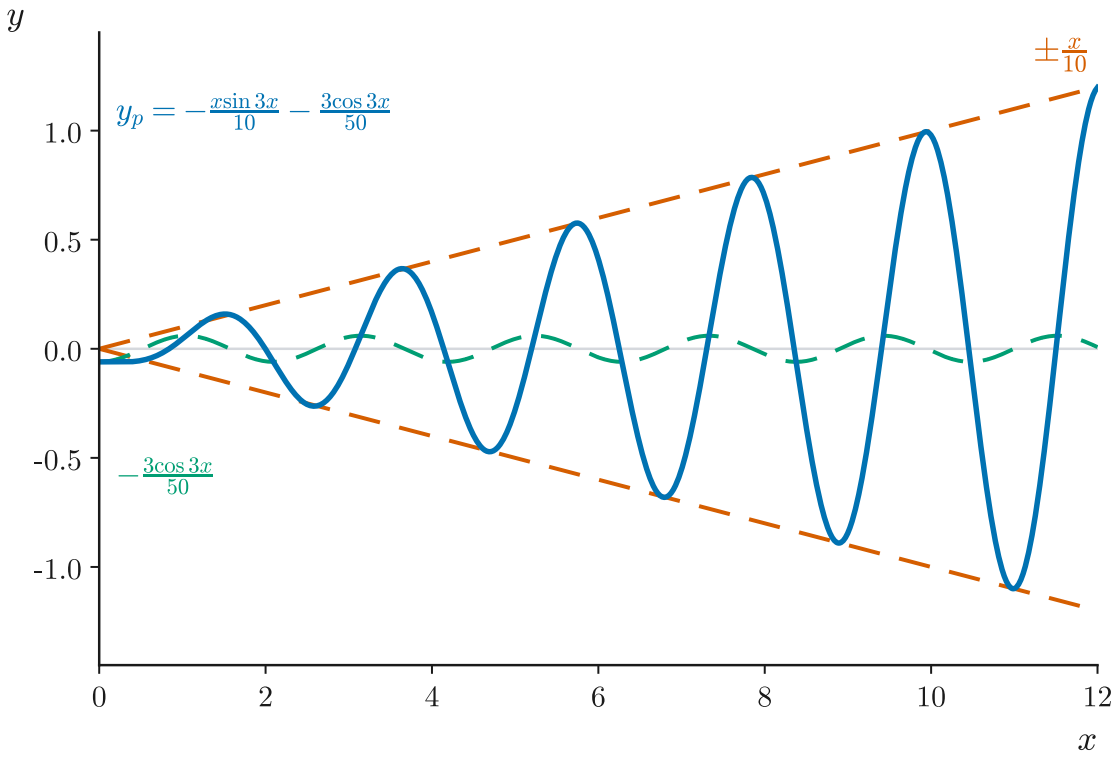
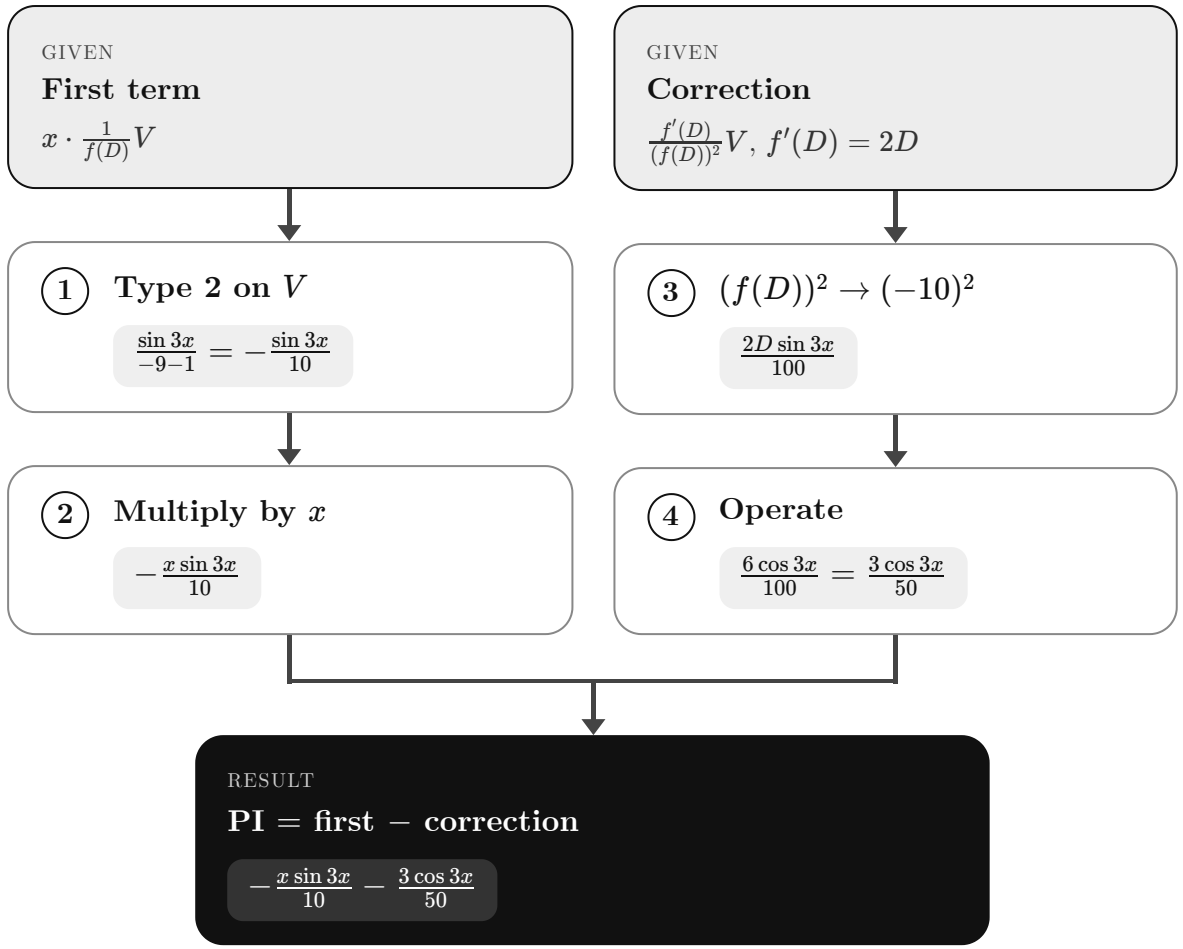


Figure G3.1. Type 5: the PI of $(D^2-1)y=x\sin 3x$ grows like $\frac{x}{10}$. ESA ★★

ADDED FOR CLARITY CW U2 P19

How to read it. The PI $-\frac{x\sin 3x}{10}-\frac{3\cos 3x}{50}$ (blue) has an amplitude that grows linearly, like the right-hand side $x\sin 3x$ scaled by $\frac{1}{f(-9)}=-\frac{1}{10}$; the lines $\pm\frac{x}{10}$ (vermilion dashed) are the envelope of the first term. The correction $-\frac{3\cos 3x}{50}$ (green dashed) is the $\frac{f'(D)}{(f(D))^2}V$ term: small, of fixed amplitude, and it dominates only near $x=0$.

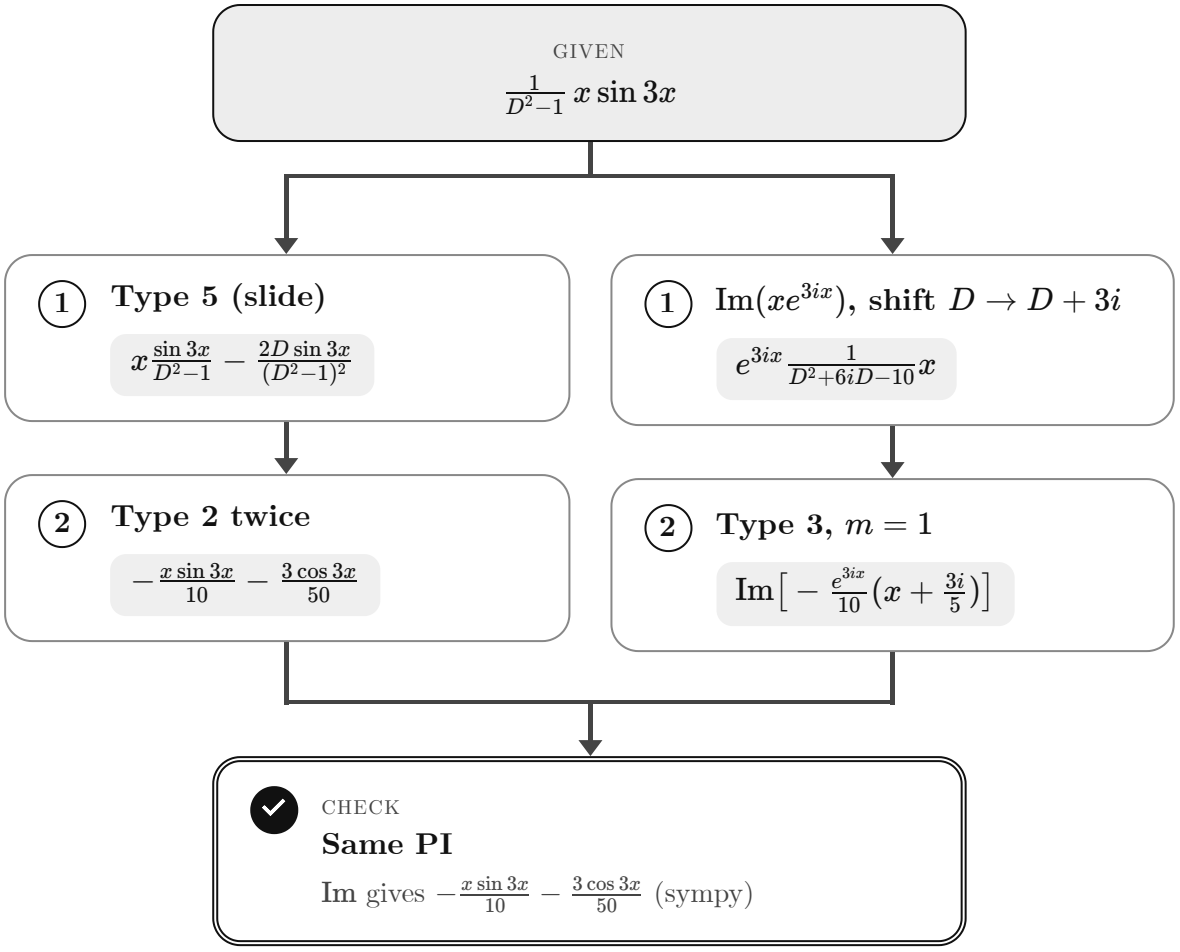
Flowcharts



Remark. Both halves are ordinary Type 2 work. Worked as W3.1; W3.2 (CW P20) needs rationalising in both halves.

Figure F3.1. The Type 5 procedure, on CW P19: $(D^2 - 1)y = x \sin 3x$ ESA ★★

S006·4 · CW U2 P19



Remark. The right-hand route is added for clarity; write the slide method (left) in the exam.

Figure F3.2. Two routes to the PI of $x\sin 3x$, one answer ESA ★

ADDED FOR CLARITY S006·4 · CW U2 P19

Type 5 formula or complex exponential?

	Type 5 formula (slide)	**Im/Re of xe^{iax}**
Steps	two Type 2 computations	one shift, one Type 3 series with complex numbers
Needs	$f'(D)$ and $(f(D))^2$	$f(D+ia)$ and complex division
Best for	the exam (it is the taught rule)	checking the answer
CW P19	$-\frac{x\sin 3x}{10} - \frac{3\cos 3x}{50}$	same
CW P20	$\frac{x(\cos 2x - 5\sin 2x)}{52} - \frac{73\sin 2x}{1352} - \frac{5\cos 2x}{169}$	same

ADDED FOR CLARITY S006·4 · CW U2 P19 · CW U2 P20

Important points and why

1. Type 5: $\frac{1}{f(D)}(xV) = x\frac{1}{f(D)}V - \frac{f'(D)}{(f(D))^2}V$. Two Type 2 computations: first $\frac{V}{f(D)}$, then $\frac{f'(D)V}{(f(D))^2}$. ESA ★★ S006·4 · T006 00:01:28 · CW U2 P19 · CW U2 P20
2. In $\frac{f'(D)}{(f(D))^2}V$, replace D^2 by $-a^2$ in $(f(D))^2$ after squaring, then rationalise any leftover D. ESA ★★ ADDED FOR CLARITY CW U2 P20
3. Check with the complex shortcut: $x\sin ax = \mathrm{Im}(xe^{iax})$, Type 4 with $a \to ia$ and $V = x$. ESA ★ ADDED FOR CLARITY CW U2 P19

Why is there a correction term $-\frac{f'(D)}{(f(D))^2}V$?
Multiplying by x does not commute with D: $D(xU) = xDU + U$. So $f(D)$ applied to $x\frac{V}{f(D)}$ gives xV plus the extra $f'(D)\frac{V}{f(D)}$; the correction term removes that extra (E3.2).
ADDED FOR CLARITY S006·4

Exam trap. $\frac{x}{f(D)}V$ is $x \cdot \left(\frac{1}{f(D)}V\right)$, not $\frac{1}{f(D)}(xV)$ (that is what you are trying to find).
ADDED FOR CLARITY S006·4

Exam trap. Square $f(D)$ before substituting $D^2 = -a^2$ is fine, but keep the cross term: $(2 - 5D)^2 = 4 - 20D + 25D^2$, not $4 + 25D^2$. ADDED FOR CLARITY CW U2 P20

Worked examples

W3.1 · $(D^2 - 1)y = x\sin 3x$ (CW P19) — ESA ★★

GIVEN Solve $(D^2 - 1)y = x\sin 3x$.

STEP 1 AE $m^2 - 1 = 0$: CF $= c_1e^x + c_2e^{-x}$. $f(D) = D^2 - 1$, $f'(D) = 2D$, $V = \sin 3x$.

STEP 2 $\frac{x}{f(D)}\sin 3x = x\frac{\sin 3x}{-9-1} = -\frac{x\sin 3x}{10}$.

STEP 3 $\frac{f'(D)}{(f(D))^2}\sin 3x = \frac{2D\sin 3x}{(-10)^2} = \frac{6\cos 3x}{100} = \frac{3\cos 3x}{50}$.

STEP 4 PI $= -\frac{x\sin 3x}{10} - \frac{3\cos 3x}{50}$.

STEP 5 Check (E3.3): $\mathrm{Im}\left[e^{3ix}\frac{1}{D^2+6iD-10}x\right] = \mathrm{Im}\left[-\frac{e^{3ix}}{10}\left(x + \frac{6i}{10}\right)\right]$, the same.

ANSWER $$\boxed{y = c_1e^x + c_2e^{-x} - \frac{x\sin 3x}{10} - \frac{3\cos 3x}{50}}$$

CHECK CW answer agrees; sympy dsolve and checkodesol agree. Figure G3.1.

CW U2 P19 · S006·4

W3.2 · $y''-5y'+6y=x\cos 2x$ (CW P20, homework) ESA ★★

GIVEN Solve $y''-5y'+6y=x\cos 2x$.

STEP 1 AE $(m-2)(m-3)=0$: CF $=c_1e^{2x}+c_2e^{3x}$. $f(D)=D^2-5D+6$, $f'(D)=2D-5$.

STEP 2 $\frac{\cos 2x}{f(D)}$: $D^2\to -4$: $\frac{\cos 2x}{2-5D}=\frac{(2+5D)\cos 2x}{4-25D^2}=\frac{2\cos 2x-10\sin 2x}{104}$; times x.

STEP 3 $(f(D))^2\to(2-5D)^2=4-20D+25D^2\to -96-20D=-4(24+5D)$.

STEP 4 $\frac{(2D-5)\cos 2x}{-4(24+5D)}=-\frac{(2D-5)(24-5D)\cos 2x}{4(576+100)}=-\frac{(73D-80)\cos 2x}{2704}=\frac{146\sin 2x+80\cos 2x}{2704}$.

STEP 5 PI $=\frac{x(2\cos 2x-10\sin 2x)}{104}-\frac{146\sin 2x+80\cos 2x}{2704}$.

ANSWER
$$\boxed{y=c_1e^{2x}+c_2e^{3x}+\frac{x(\cos 2x-5\sin 2x)}{52}-\frac{73\sin 2x}{1352}-\frac{5\cos 2x}{169}}$$

CHECK CW answer agrees (it is the same, unreduced); sympy dsolve and checkodesol agree. By E3.3: $\mathrm{Re}\left[e^{2ix}\left(\frac{(1+5i)x}{52}-\frac{5}{169}+\frac{73i}{1352}\right)\right]$, the same.

CW U2 P20 · S006·4

SHORT-NOTE RECAP

Type 5: $X=xV$. $\frac{1}{f(D)}(xV)=x\frac{1}{f(D)}V-\frac{f'(D)}{(f(D))^2}V$, both parts by Type 2. CW P19: $\frac{x\sin 3x}{D^2-1}=-\frac{x\sin 3x}{10}-\frac{3\cos 3x}{50}$. CW P20: $\frac{x\cos 2x}{D^2-5D+6}=\frac{x(\cos 2x-5\sin 2x)}{52}-\frac{73\sin 2x}{1352}-\frac{5\cos 2x}{169}$. Check: $x\sin ax=\mathrm{Im}(xe^{iax})$, then Type 4 with $V=x$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $\frac{1}{D}$ = integration without a constant	E1.1, E2.2
$f(D)$	Operator polynomial	Left-hand side operator of $f(D)y = X$	E1.1, E3.1
$f(D+a)$	Shifted operator	$f(D)$ with every D replaced by $D+a$	E1.1, E1.2
$f'(D)$	f dash of D	Derivative of the polynomial $f(D)$ with respect to D	E3.1, E3.2
a	Exponent coefficient	Coefficient of x in e^{ax}; in Type 1 and Type 2 also the frequency of $\sin ax$ (clash)	E1.1
b	Frequency of V	$V = \sin bx$ or $\cos bx$; $D^2 \to -b^2$ after the shift	E1.3
V	Remaining factor	$V(x)$: what is left after e^{ax} (Type 4) or x (Type 5) is split off	E1.1, E3.1
$\varphi(D)$	Phi of D	$f(D+a) = f(a)[1 \pm \varphi(D)]$, only positive powers of D	E2.1
b_0	Constant of $f(D+a)$	Equals $f(a)$; taken out before the series	E2.1
$\psi(D)$	Psi of D	$f(D) = (D-a)^k \psi(D)$ with $\psi(a) \neq 0$	E1.4
Im, Re	Imaginary, real part	$x \sin ax = \mathrm{Im}(xe^{iax})$, $x \cos ax = \mathrm{Re}(xe^{iax})$	E3.3
PI_1, PI_2	Partial PIs	PIs of the separate terms of X	E2.4

B Sources, coverage and gaps

Source	Path	Use
Deck 006 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.*	15 slides; content on 3–14 (every equation read from the rendered images)
Transcript T006 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.*	17 min; slides 4–9 and one extra example
Transcript T006b (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.*	15.5 min; 00:09:55–00:15:25 Type 4 and slides 10–14 (00:00:00–00:09:55 is Type 3, used by the class 005 pack as T006a)
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Unit 2 problems 17–20 (all solved in full; all four printed answers are right, P20 is left unsimplified)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Jan-24 set 3 2c, via the shared ESA map
Video 006 (.mp4, both parts)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_*.mp4	Not opened: every slide equation is readable and no video trigger applies

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences; 7 distinct / 9 occurrences are PI by operator rules). Type 4 is asked once: Jan-24 set 3 2c, $(D^3 - 2D^2 + D)y = x^2e^{-x} + \sin^2 x$ (6 marks), which is also CW P18 (solved as W2.3). Type 5 (xV) is never asked on its own; it is computed only in CW P19 and P20. Stars: the Type 4 rule and $x^m e^{ax}$ ★★ (one ESA question plus CW P17, P18); Type 5 ★★ (CW only); the complex-exponential shortcut ★ (added for clarity).

Known gaps, conflicts and corrections

1. Slide 006·9 (and the lecture, T006 00:07:16) integrates $x \sin x$ where it should integrate $x \cos x$: after the shift the PI is $e^x \frac{1}{D^2}(x \cos x)$, and $\int x \cos x \, dx = x \sin x + \cos x$, but the slide writes $e^x \frac{1}{D} \int x \sin x \, dx$. Its boxed answer $e^x[-(x \sin x + \cos x) - \cos x] = -e^x(x \sin x + 2 \cos x)$ is $e^x \frac{1}{D^2}(x \sin x)$, and substituting it back does not give $xe^x \cos x$ (sympy). The correct PI is $e^x(2 \sin x - x \cos x)$ (W2.2, checked by dsolve and by the Type 5 formula).
2. Slide 006·5 writes the CF as $e^x(c_1 \cos \sqrt{3}x + c_1 \cos \sqrt{3}x)$; it is $e^x(c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x)$, as the lecture says (T006 00:03:06).
3. Slide 006·3 reads "$f(D)y = x$ When $X = e^{ax}V$"; it means $f(D)y = X$ (capital X).

4. No slide and no lecture works a Type 5 (xV) example; the rule is only stated (slide 006·4, T006 00:01:28). Its derivation and the complex-exponential shortcut ($x\sin ax = \operatorname{Im} xe^{iax}$) are added for clarity; CW P19 and P20 are solved both ways.
5. T006's closing sentence (00:16:51) says "this is how we find the particular integral when the given function is x into V"; the lecture was on $e^{ax}V$ (Type 4). T006b 00:11:57 says "e^{ax} is taken as e^{x}" for x^2e^{3x}; it is e^{3x} ($a=3$), as the next sentence uses. T006b 00:09:55 calls Type 4 "a combination of case four plus case three and case two": after the shift the rest is done by Type 2 or Type 3 (00:10:53).
6. The example $(D^2-3D+2)y = xe^{3x}+\sin 2x$ (T006 00:09:45–00:16:39) is on no slide; it is worked here from the lecture (W2.4, flagged transcript only), with the general solution the lecture leaves as "PI_1 + PI_2".
7. All other slide answers were recomputed with sympy (dsolve, checkodesol): 006·6 $\frac{e^x\cos x}{2}$, 006·12 $\frac{e^{3x}}{11}(x^2-\frac{12x}{11}+\frac{50}{121})$ and 006·14 $-e^x\sin x$ are right; the AE roots on slides 006·5 ($1\pm\sqrt{3}i$), 006·7 and 006·13 ($1,1$) and 006·11 ($\pm\sqrt{2}i$) agree with their own CFs. Slides 006·5–6 and 006·7–9 stop at "Thus, $y=y_c+y_p$": the general solutions are written out here.
8. CW answers recomputed: P17 $\frac{e^{2x}(7\cos x+4\sin x)}{130}$, P18, P19 $-\frac{x\sin 3x}{10}-\frac{3\cos 3x}{50}$ and P20 are all right. P20's printed form $\frac{x}{104}(2\cos 2x-10\sin 2x)+\frac{1}{2704}(-146\sin 2x-80\cos 2x)$ looks odd only because it is not reduced: it equals $\frac{x(\cos 2x-5\sin 2x)}{52}-\frac{73\sin 2x}{1352}-\frac{5\cos 2x}{169}$ (sympy agrees with both forms).
9. ESA Jan-24 set 3 2c is the same question as CW P18 and as slide 004·17 (its $\sin^2 x$ part). The class 004 pack found that slide 004·17's boxed PI for the $\sin^2 x$ part drops a factor 2; the correct part, used here, is $\frac{x}{2}-\frac{\cos 2x}{25}+\frac{3\sin 2x}{100}$, which agrees with CW P18.
10. Transcript recognition errors were corrected silently: "e par ax", "e bar ax", "e par ex", "epo x" → e^{ax} or e^{x}; "into b", "of b", "d plus a of b" → V; "replacing b by d plus a" → replacing D by $D+a$; "f dash d" → $f'(D)$; "c even" → c_1; "psi root three x" → $\sin\sqrt{3}x$; "cos 6" → $\cos x$; "minus 2t" → $-2D$; "particular interval", "particular integrity" → particular integral; "3 into 39" → $3\cdot 3=9$; "5d", "5 of d" → $\varphi(D)$; "1 by 4 t" → $\frac{1}{40}$; "sine to x" → $\sin 2x$; "thirst" → first; "Yc plus YPC" → y_c+y_p; repeated "PES University" (T006 00:11:49, the video's watermark) dropped.

Coverage

Source	Citations	Slides cited (slide × citations)
S006	85	3×1, 4×25, 5×4, 6×10, 7×2, 8×7, 9×7, 10×6, 11×6, 12×8, 13×2, 14×7
CW	32	
ESA	8	
T006	48	
T006b	23	

Slides 1, 2 and 15 are title and closing slides; slide 3 is the class content and slide 4 the rules for Types (iv) and (v). T006 (17 min) covers slides 4–9 and adds one example on no slide (W2.4, $(D^2-3D+2)y=xe^{3x}+\sin 2x$). Slides 10–14 (the newer "Type 4: $X=e^{ax}V$" format with two problems) are explained in T006b from 00:09:55 to the end; the first 9 min 55 s of that file are Type 3 and are cited as T006a in the class 005 pack, so the same file has two tags in the unit

master (T006a there, T006b here) and only its Type 4 part is cited here. No slide or lecture works a Type 5 example: its formula is stated (slide 006·4, T006 00:01:28) and the lecturer says “in this session, we are solving problems only on type 4”; CW P19 and P20 supply the Type 5 problems. Bucket numbers are not contiguous (the decks say SESSION 1–11; 007, 008, 010, 012 and 013 do not exist; 009 = session 7, 011 = sessions 8 and 9, 014 = 10, 015 = 11); this deck is SESSION 6.